

Time Series — Solutions

Practice Exam — Week 3: Backshift, Wold Decomposition, Causality & Invertibility
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Problem 1 (5 points)

Statement.

(a) Define the backshift operator B and, for an ARMA(p, q) process written $\Phi(B)X_t = \Theta(B)\varepsilon_t$, write down the characteristic polynomials $\Phi(z)$ and $\Theta(z)$. State the root criteria (in terms of the unit circle) for (i) stationarity/causality and (ii) invertibility. (2 pts)

(b) Consider the AR(2) process

$$X_t = \frac{2}{3}X_{t-1} - \frac{1}{6}X_{t-2} + \varepsilon_t.$$

Write its AR polynomial $\Phi(z)$, find the roots, and decide whether the process is stationary. (2 pts)

(c) Is the process of part (b) invertible? Justify in one sentence (no computation needed). (1 pt)

Solution.

(a) The backshift operator B shifts a series one step into the past, $BX_t = X_{t-1}$, and iterating $B^k X_t = X_{t-k}$ (with $B^0 = I$). For an ARMA(p, q) written $\Phi(B)X_t = \Theta(B)\varepsilon_t$ the characteristic polynomials are

$$\Phi(z) = 1 - \phi_1 z - \dots - \phi_p z^p, \quad \Theta(z) = 1 + \theta_1 z + \dots + \theta_q z^q$$

(coefficients matching the chosen sign convention). The root criteria are:

- **Stationarity / causality:** all roots of $\Phi(z)$ lie *outside* the unit circle, $|z| > 1$;
- **Invertibility:** all roots of $\Theta(z)$ lie *outside* the unit circle, $|z| > 1$.

(b) In backshift form $(1 - \frac{2}{3}B + \frac{1}{6}B^2)X_t = \varepsilon_t$, so

$$\Phi(z) = 1 - \frac{2}{3}z + \frac{1}{6}z^2.$$

Solve $\Phi(z) = 0$. Multiplying by 6 gives $z^2 - 4z + 6 = 0$, hence

$$z = \frac{4 \pm \sqrt{16 - 24}}{2} = 2 \pm \sqrt{-2} = 2 \pm \sqrt{2}i, \quad |z| = \sqrt{2^2 + (\sqrt{2})^2} = \sqrt{4 + 2} = \sqrt{6} \approx 2.449.$$

Both roots have modulus $\sqrt{6} > 1$, i.e. lie outside the unit circle, so the process is **stationary** (causal). (One may also check the AR(2) triangle with $\phi_1 = \frac{2}{3}$, $\phi_2 = -\frac{1}{6}$: $\phi_1 + \phi_2 = \frac{1}{2} < 1$, $\phi_2 - \phi_1 = -\frac{5}{6} < 1$, $|\phi_2| = \frac{1}{6} < 1$.)

(c) **Yes, invertible.** It is a pure AR process ($\Theta \equiv 1$, a finite polynomial with no roots to test), so $\varepsilon_t = \Phi(B)X_t$ is already a finite expression in present and past X : every AR is automatically invertible.

Problem 2 (5 points)

Statement.

For the ARMA(2, 1) process

$$(1 - 0.7B + 0.1B^2)X_t = (1 + 0.5B)\varepsilon_t,$$

(a) factor the AR polynomial $\Phi(z) = 1 - 0.7z + 0.1z^2$, locate its roots, and confirm that the process is causal. (2 pts)

(b) Using a partial-fraction expansion of the transfer function $G(z) = \Theta(z)/\Phi(z)$, find a closed form for the $\text{MA}(\infty)$ weights ψ_j in $X_t = \sum_{j \geq 0} \psi_j \varepsilon_{t-j}$, and report ψ_0, ψ_1, ψ_2 . (3 pts)

Solution.

(a) $\Phi(z) = 1 - 0.7z + 0.1z^2 = 0.1(z^2 - 7z + 10) = 0.1(z - 2)(z - 5)$, so the AR roots are $z = 2$ and $z = 5$. Both satisfy $|z| > 1$, hence all roots of Φ are outside the unit circle and the process is **causal** (stationary). Also $\Theta(z) = 1 + 0.5z$ has root $z = -2$, $|z| = 2 > 1$, so it is invertible as well (not asked, but it guarantees a well-defined G).

(b) Write $X_t = G(B)\varepsilon_t$ with

$$G(z) = \frac{1 + 0.5z}{0.1(z - 2)(z - 5)}.$$

Partial-fraction the rational factor:

$$\frac{1}{(z - 2)(z - 5)} = \frac{A}{z - 2} + \frac{B}{z - 5}, \quad A = \frac{1}{2 - 5} = -\frac{1}{3}, \quad B = \frac{1}{5 - 2} = \frac{1}{3}.$$

Convert each pole to a power series convergent on (and inside) the unit disk, using $|z| < 2 < 5$:

$$\begin{aligned} \frac{-1}{z - 2} &= \frac{1}{2} \frac{1}{1 - z/2} = \frac{1}{2} \sum_{j \geq 0} \left(\frac{1}{2}\right)^j z^j = \sum_{j \geq 0} \left(\frac{1}{2}\right)^{j+1} z^j, \\ \frac{1}{z - 5} &= -\frac{1}{5} \frac{1}{1 - z/5} = -\frac{1}{5} \sum_{j \geq 0} \left(\frac{1}{5}\right)^j z^j = -\sum_{j \geq 0} \left(\frac{1}{5}\right)^{j+1} z^j. \end{aligned}$$

Hence, collecting $\frac{1}{(z - 2)(z - 5)} = \frac{A(-1)}{2 - z} \dots$ explicitly,

$$\frac{1}{(z - 2)(z - 5)} = \left(-\frac{1}{3}\right) \frac{1}{z - 2} + \frac{1}{3} \frac{1}{z - 5} = \sum_{j \geq 0} \left[\frac{1}{3} \left(\frac{1}{2}\right)^{j+1} - \frac{1}{3} \left(\frac{1}{5}\right)^{j+1} \right] z^j.$$

Now multiply by $\frac{1 + 0.5z}{0.1} = 10(1 + \frac{1}{2}z)$. Write $H(z) = \sum_{j \geq 0} h_j z^j$ with

$$h_j = 10 \left[\frac{1}{3} \left(\frac{1}{2}\right)^{j+1} - \frac{1}{3} \left(\frac{1}{5}\right)^{j+1} \right] = \frac{5}{3} \left(\frac{1}{2}\right)^j - \frac{2}{3} \left(\frac{1}{5}\right)^j,$$

so that $G(z) = (1 + \frac{1}{2}z)H(z)$ gives $\psi_0 = h_0$ and $\psi_j = h_j + \frac{1}{2}h_{j-1}$ for $j \geq 1$. Computing the coefficients of $(1/2)^j$ and $(1/5)^j$:

$$\begin{aligned} \text{coeff of } \left(\frac{1}{2}\right)^j &: \quad \frac{5}{3} + \frac{1}{2} \cdot \frac{5}{3} \cdot 2 = \frac{5}{3} + \frac{5}{3} = \frac{10}{3}, \\ \text{coeff of } \left(\frac{1}{5}\right)^j &: \quad -\frac{2}{3} + \frac{1}{2} \cdot \left(-\frac{2}{3}\right) \cdot 5 = -\frac{2}{3} - \frac{5}{3} = -\frac{7}{3}, \end{aligned}$$

(using $(\frac{1}{2})^{j-1} = 2(\frac{1}{2})^j$ and $(\frac{1}{5})^{j-1} = 5(\frac{1}{5})^j$). Therefore

$$\boxed{\psi_j = \frac{10}{3} \left(\frac{1}{2}\right)^j - \frac{7}{3} \left(\frac{1}{5}\right)^j, \quad j \geq 0}$$

which already gives $\psi_0 = \frac{10}{3} - \frac{7}{3} = 1$. In particular

$$\psi_0 = 1, \quad \psi_1 = \frac{10}{3} \cdot \frac{1}{2} - \frac{7}{3} \cdot \frac{1}{5} = \frac{5}{3} - \frac{7}{15} = \frac{6}{5} = 1.2, \quad \psi_2 = \frac{10}{3} \cdot \frac{1}{4} - \frac{7}{3} \cdot \frac{1}{25} = \frac{5}{6} - \frac{7}{75} = \frac{37}{50} = 0.74.$$

Check (coefficient matching). From $(1 - 0.7z + 0.1z^2) \sum_j \psi_j z^j = 1 + 0.5z$: $\psi_0 = 1$; $\psi_1 - 0.7\psi_0 = 0.5 \Rightarrow \psi_1 = 1.2$; and $\psi_j = 0.7\psi_{j-1} - 0.1\psi_{j-2}$ for $j \geq 2$, giving $\psi_2 = 0.7(1.2) - 0.1(1) = 0.74$, matching exactly.

Problem 3 (5 points)

Statement.

Let $\{X_t\}$ be the causal AR(2) process

$$X_t = \frac{5}{6}X_{t-1} - \frac{1}{6}X_{t-2} + \varepsilon_t.$$

(a) Show that the autocovariance sequence satisfies the homogeneous recurrence $\gamma_\tau = \frac{5}{6}\gamma_{\tau-1} - \frac{1}{6}\gamma_{\tau-2}$ for $\tau \geq 1$, and obtain the boundary relation at $\tau = 0$ involving σ^2 . (2 pts)

(b) Solve for γ_0, γ_1 and hence give the closed-form expression of γ_τ for all $\tau \geq 0$. (3 pts)

Hint: the right-hand side of the model equation only contributes through $\mathbb{E}[\varepsilon_t X_t] = \sigma^2$ (causality).

Solution.

(a) The model is $X_t - \frac{5}{6}X_{t-1} + \frac{1}{6}X_{t-2} = \varepsilon_t$, with $\Phi(z) = 1 - \frac{5}{6}z + \frac{1}{6}z^2 = \frac{1}{6}(z-2)(z-3)$ (roots 2, 3 outside the unit circle $\Rightarrow$ causal). Multiply the model by $X_{t-\tau}$ and take expectations. Since the process is causal, $X_{t-\tau}$ is a one-sided combination of $\varepsilon_{t-\tau}, \varepsilon_{t-\tau-1}, \dots$, so $\mathbb{E}[\varepsilon_t X_{t-\tau}] = 0$ for $\tau \geq 1$ and $\mathbb{E}[\varepsilon_t X_t] = \psi_0 \sigma^2 = \sigma^2$. Hence, for $\tau \geq 1$,

$$\gamma_\tau - \frac{5}{6}\gamma_{\tau-1} + \frac{1}{6}\gamma_{\tau-2} = \mathbb{E}[\varepsilon_t X_{t-\tau}] = 0 \implies \gamma_\tau = \frac{5}{6}\gamma_{\tau-1} - \frac{1}{6}\gamma_{\tau-2},$$

(using $\gamma_{-1} = \gamma_1$ when $\tau = 1$). The boundary relation at $\tau = 0$ keeps the noise term:

$$\gamma_0 - \frac{5}{6}\gamma_1 + \frac{1}{6}\gamma_2 = \mathbb{E}[\varepsilon_t X_t] = \sigma^2.$$

(b) The recurrence at $\tau = 1, 2$ (with $\gamma_{-1} = \gamma_1$) gives the autocorrelations via Yule–Walker:

$$\rho_1 = \frac{\phi_1}{1 - \phi_2} = \frac{5/6}{1 + 1/6} = \frac{5/6}{7/6} = \frac{5}{7}, \quad \rho_2 = \phi_1 \rho_1 + \phi_2 = \frac{5}{6} \cdot \frac{5}{7} - \frac{1}{6} = \frac{25}{42} - \frac{7}{42} = \frac{18}{42} = \frac{3}{7}.$$

Plug into the $\tau = 0$ relation, $\gamma_0(1 - \phi_1 \rho_1 - \phi_2 \rho_2) = \sigma^2$:

$$1 - \frac{5}{6} \cdot \frac{5}{7} - \left(-\frac{1}{6}\right) \frac{3}{7} = 1 - \frac{25}{42} + \frac{3}{42} = 1 - \frac{22}{42} = \frac{20}{42} = \frac{10}{21}, \quad \gamma_0 = \frac{21}{10} \sigma^2.$$

Therefore $\gamma_1 = \rho_1 \gamma_0 = \frac{5}{7} \cdot \frac{21}{10} \sigma^2 = \frac{3}{2} \sigma^2$ and $\gamma_2 = \rho_2 \gamma_0 = \frac{3}{7} \cdot \frac{21}{10} \sigma^2 = \frac{9}{10} \sigma^2$.

The homogeneous recurrence has characteristic equation Φ , whose roots correspond to the reciprocal roots $\frac{1}{2}, \frac{1}{3}$, so the general solution is $\gamma_\tau = c_1 \left(\frac{1}{2}\right)^\tau + c_2 \left(\frac{1}{3}\right)^\tau$. Match $\gamma_0 = \frac{21}{10} \sigma^2$ and $\gamma_1 = \frac{3}{2} \sigma^2$:

$$\begin{aligned} c_1 + c_2 &= \frac{21}{10} \sigma^2, \\ \frac{1}{2}c_1 + \frac{1}{3}c_2 &= \frac{3}{2} \sigma^2. \end{aligned}$$

Subtracting $\frac{1}{3} \times (\text{first})$ from the second, $\frac{1}{6}c_1 = \left(\frac{3}{2} - \frac{21}{30}\right) \sigma^2 = \frac{45-21}{30} \sigma^2 = \frac{24}{30} \sigma^2$, so $c_1 = \frac{24}{5} \sigma^2$ and $c_2 = \frac{21}{10} \sigma^2 - \frac{24}{5} \sigma^2 = -\frac{27}{10} \sigma^2$. Hence

$$\boxed{\gamma_\tau = \left[\frac{24}{5} \left(\frac{1}{2}\right)^\tau - \frac{27}{10} \left(\frac{1}{3}\right)^\tau \right] \sigma^2, \quad \tau \geq 0}$$

and $\gamma_{-\tau} = \gamma_\tau$. (Check: $\tau = 0 \rightarrow \frac{48-27}{10} = \frac{21}{10}$; $\tau = 1 \rightarrow \frac{24}{10} - \frac{27}{30} = \frac{72-27}{30} = \frac{45}{30} = \frac{3}{2}$; $\tau = 2 \rightarrow \frac{24}{20} - \frac{27}{90} = \frac{6}{5} - \frac{3}{10} = \frac{9}{10}$, all matching.)

Independent check (via ψ -weights): $\psi_j = 2\left(\frac{1}{2}\right)^j - \left(\frac{1}{3}\right)^j$ solves $\psi_j = \frac{5}{6}\psi_{j-1} - \frac{1}{6}\psi_{j-2}$ with $\psi_0 = 1, \psi_1 = \frac{5}{6}$, and $\sigma^2 \sum_{j \geq 0} \psi_j \psi_{j+\tau}$ reproduces $\frac{21}{10}, \frac{3}{2}, \frac{9}{10} \dots$ exactly.

Problem 4 (5 points)

Statement.

(Revision of Week 2 + this week.) Let $X_t = \varepsilon_t - \theta \varepsilon_{t-1}$ be an MA(1) with $\theta = \frac{1}{2}$.

- (a) Compute the autocovariance sequence γ_τ for all τ and the autocorrelation ρ_1 . (1.5 pts)
- (b) Prove that any MA(1) of the form $\varepsilon_t - \theta\varepsilon_{t-1}$ satisfies $|\rho_1| \leq \frac{1}{2}$, and state qualitatively how the PACF of an MA(1) behaves (cut off or tail off?). (1.5 pts)
- (c) Is $\{X_t\}$ invertible? Then exhibit a different MA(1) process $Y_t = \eta_t - \vartheta\eta_{t-1}$ (with an appropriately chosen white-noise variance) that has exactly the same autocovariance sequence as $\{X_t\}$, and explain which of the two is the invertible representation. (2 pts)

Solution.

(a) With $X_t = \varepsilon_t - \frac{1}{2}\varepsilon_{t-1}$, bilinearity and $\text{Cov}(\varepsilon_s, \varepsilon_u) = \sigma^2 \mathbf{1}_{\{s=u\}}$ give

$$\gamma_\tau = \begin{cases} (1 + \theta^2)\sigma^2 = (1 + \frac{1}{4})\sigma^2 = \frac{5}{4}\sigma^2, & \tau = 0, \\ -\theta\sigma^2 = -\frac{1}{2}\sigma^2, & \tau = \pm 1, \\ 0, & |\tau| \geq 2, \end{cases} \quad \rho_1 = \frac{-\theta}{1 + \theta^2} = \frac{-1/2}{5/4} = -\frac{2}{5}.$$

(b) For a general MA(1), $\rho_1 = \frac{-\theta}{1 + \theta^2}$, so

$$|\rho_1| = \frac{|\theta|}{1 + \theta^2} \leq \frac{1}{2} \iff 2|\theta| \leq 1 + \theta^2 \iff 0 \leq \theta^2 - 2|\theta| + 1 = (|\theta| - 1)^2,$$

which always holds (equality iff $|\theta| = 1$). Equivalently, by AM–GM $|\theta| \leq \frac{1}{2}(1 + \theta^2)$. So $|\rho_1| \leq \frac{1}{2}$.

PACF behaviour: for an MA(1) the **PACF tails off** (decays geometrically in magnitude with alternating sign, never cutting to exactly 0), while it is the ACF that cuts off after lag 1. (This is the diagnostic mirror of the AR(1), whose PACF cuts off after lag 1 and ACF tails off.)

(c) **Invertible:** $\Theta(z) = 1 - \frac{1}{2}z$ has root $z = 1/\theta = 2$ with $|2| > 1$ (outside the unit circle), so $\{X_t\}$ is invertible.

A different MA(1) with the same ACVS is the *reciprocal-root twin*: take coefficient $\vartheta = 1/\theta = 2$ and white-noise variance $\sigma_\eta^2 = \theta^2\sigma^2 = \frac{1}{4}\sigma^2$, i.e.

$$Y_t = \eta_t - 2\eta_{t-1}, \quad \text{Var}(\eta_t) = \frac{1}{4}\sigma^2.$$

Its autocovariances are

$$\gamma_0^Y = (1 + \vartheta^2)\sigma_\eta^2 = (1 + 4) \cdot \frac{1}{4}\sigma^2 = \frac{5}{4}\sigma^2, \quad \gamma_{\pm 1}^Y = -\vartheta\sigma_\eta^2 = -2 \cdot \frac{1}{4}\sigma^2 = -\frac{1}{2}\sigma^2,$$

and $\gamma_\tau^Y = 0$ for $|\tau| \geq 2$ — *identical* to $\{X_t\}$ in part (a). So the two are indistinguishable from second-order structure alone (MA non-identifiability). The **invertible** representation is $\{X_t\}$, whose MA root $z = 2$ lies *outside* the unit circle; the twin $\{Y_t\}$ has root $z = \vartheta^{-1} = \frac{1}{2}$ *inside* the circle and is therefore *not* invertible. Imposing invertibility selects $\{X_t\}$ as the unique model.